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# KESTREL

The drone security analyst that never blinks

## Approach, Decisions and Results

A drone security analyst built around one constraint: you cannot look at every frame

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8/8 scenarios correct · 100 tests passing

Every figure in this document is generated from a measured run. Re-running the evaluation suite regenerates the tables; no number is transcribed by hand.

FlytBase AI Engineer assignment

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## 1 The problem, restated

The brief asks for an agent that processes telemetry and video, logs objects with context, raises alerts, and indexes frames for retrieval. Read literally, that is a week of work. The interesting version of the problem only appears once you count.

A docked patrol drone at a modest 2fps produces about **57,600 frames per eight-hour shift**. The free inference tier permits roughly **40 requests per minute**. Captioning every frame would take about twenty-four hours of wall-clock per shift and cost more than the guard it replaces.

And almost every one of those frames shows the same empty yard.

### The organising constraint

The hard problem is not detection. It is deciding *what deserves attention*, carrying enough context to know, and being right often enough that an operator still reads the alerts in week three. Every architectural decision in KESTREL follows from that.

## 2 Approach

KESTREL is structured as a **cost cascade**: each tier only runs when the tier before it has earned the spend.

| Tier | What it does                                             | Cost      |
|------|----------------------------------------------------------|-----------|
| 0    | Gate — is this frame worth spending anything on?         | free, CPU |
| 1    | Detect — YOLO11 on the local GPU                         | ~12 ms    |
| 1.5  | Track — ByteTrack, persistent identity                   | ~free     |
| 2    | Project and embed — pixel to world, vectors for re-ID    | cheap     |
| 3    | Perceive — vision model returns a structured scene graph | ~1.3 s    |
| 4    | Escalate — deep model, <i>asynchronous only</i>          | 57–84 s   |

Four decisions in that table are worth defending.

### 2.1 Tier 1.5 is not optional

Without a tracker, “a person” in frame 41 and “a person” in frame 42 are unrelated observations, and every claim about *duration* is a guess. Dwell, loitering, “the same vehicle returned” — the entire temporal vocabulary the brief’s own examples require — depends on identity persisting across frames. It is the cheapest tier and the one whose absence would quietly invalidate the most.

### 2.2 Escalation splits by the kind of doubt

The plan assumed escalation meant “call a bigger model”. Measured, the deep vision model takes 57–84 seconds. Awaiting that inside frame processing would stall the pipeline for a minute per escalation.

So escalation is split by *why* we are unsure:

- **Semantic** — we need to *see* better (occlusion, poor light). Routes to the deep model **asynchronously**; the record is written at tier-3 confidence and upgraded when the deeper answer lands.
- **Cognitive** — we need to *think* better (is this pattern a threat given the last four days?). Routes to the reasoning model over the structured scene graph in ~230 ms, on the critical path.

Most escalations are cognitive, so the common case stays fast. This design exists only because the latency was measured rather than assumed.

## 2.3 Detection runs on-device, and that is an upgrade

No hosted detector is reachable at all (§4). Detection therefore runs locally — which is the right place for it anyway: it is per-frame work, it costs no API budget, it is not rate-limited, and it keeps working with the network unplugged, which is how a drone-in-a-box operates when the site link drops.

## 2.4 The gate is the whole scalability argument

Three cheap CPU signals decide whether a frame reaches a model: perceptual-hash distance (structural change), mean pixel delta (motion the hash smooths over), and embedding distance against a rolling buffer (*same pixels, new meaning*). Context adapts sensitivity — a high-priority zone, night, or hours outside a zone’s declared schedule all lower the bar.

### A capped multiplier, learned the hard way

The context priors multiply. Over a restricted zone at night they compounded to roughly 6×, making the gate so sensitive that sensor noise tripped it and *nothing was gated at all*. The multiplier is now capped at 2.5×.

## 3 Assumptions

Stated plainly, because a system that overstates itself is harder to trust.

### What is real

The video (CC BY 4.0 footage, real pixels), the detections, the tracks, the embeddings, the captions from a live vision-language model, the rules, the memory, the retrieval, the audit chain, and every measured number in this report.

### What is simulated

**Telemetry** — there is no aircraft. Altitude, battery, wind and GPS quality are modelled, and every quantity derived from them inherits that.

**Site geometry** — zones are authored, not surveyed.

**Mission execution** — the executor integrates a flight, it does not command one.

**The fleet** — one site carries real footage; the rest are driven by a seeded generator and are flagged *simulated* in every API payload and rendered with a badge in the interface.

Further limitations: geo-projection assumes flat ground and uncalibrated camera intrinsics, so objects on elevated surfaces project long. Re-identification uses general-purpose embeddings rather than a model trained with a re-ID objective, and confuses two similar white vans more readily than a dedicated model would. Open-vocabulary detection routes through the vision model rather than a grounded detector, because the HuggingFace Hub was unreachable on the build network.

## 4 Model selection — decided by measurement

#### Four of five original model choices were wrong

Not deprecated — *unreachable*. Several models present in the provider’s catalogue return Function not found for account. There is no hosted object detector at all; the CLIP-equivalent is catalogued but unprovisioned; the reranker endpoint does not exist.

Every endpoint was probed before any dependent code was written. Twenty minutes of probing changed four model choices and one architectural decision. Had integration proceeded from the documentation, the same discoveries would have arrived tangled up in debugging half-built components.

**Table 1:** Endpoint verification. Run before any dependent code was written.

| Endpoint                                       | Latency  | Status |
|------------------------------------------------|----------|--------|
| GET /models                                    | 237 ms   | PASS   |
| chat nvidia llama-3.1-70b (failover path)      | 52008 ms | PASS   |
| chat nvidia llama-3.3-70b (unusably slow)      | 35814 ms | PASS   |
| chat structured json_schema                    | 88589 ms | PASS   |
| vlm llama-3.2-11b-vision                       | 1292 ms  | PASS   |
| vlm llama-3.2-90b-vision                       | 90063 ms | n/a    |
| embed nvclip text (unprovisioned, superseded)  | 110 ms   | n/a    |
| embed nvclip image (unprovisioned, superseded) | 95 ms    | n/a    |
| embed joint text (nemotron-embed-v1)           | 489 ms   | PASS   |
| embed joint image (nemotron-embed-v1)          | 610 ms   | PASS   |
| embed cross-modal alignment                    | 1075 ms  | PASS   |
| embed nv-embedqa-e5-v5                         | 373 ms   | PASS   |
| rerank llama-3.2-nv-rerankqa                   | 66 ms    | n/a    |
| detect nv-grounding-dino                       | 511 ms   | n/a    |
| groq llama-3.3-70b-versatile (primary)         | 271 ms   | PASS   |

**Table 2:** Every vision model in the catalogue, probed directly. Signals counts how many of three salient facts (colour, vehicle, person) the model reported for the probe image.

| Model                                   | Best latency | Signals | Status |
|-----------------------------------------|--------------|---------|--------|
| nvidia/llama-3.1-nemotron-nano-v1-8b-v1 | 0.86 s       | 2/3     | PASS   |
| meta/llama-3.2-11b-vision-instruct      | 1.28 s       | 3/3     | PASS   |
| nvidia/nemotron-nano-12b-v2-v1          | 29.21 s      | 3/3     | PASS   |
| meta/llama-3.2-90b-vision-instruct      | 57.60 s      | 3/3     | PASS   |
| nvidia/vila                             | —            | —       | 404    |
| microsoft/phi-3-vision-128k-instruct    | —            | —       | 404    |
| microsoft/kosmos-2                      | —            | —       | 404    |
| nvidia/neva-22b                         | —            | —       | 404    |
| adept/fuyu-8b                           | —            | —       | 404    |

### Catalogued is not reachable

**4 / 9** catalogued vision models were actually callable. The rest return **Function not found for account**. A model appearing in a provider’s catalogue is not evidence that you can call it, and probing first changed four of five original model choices.

## 4.1 Why the 11B and not the faster 8B

The 8B model is 0.42 s faster and **missed the person** in the probe image. For a security system, a false negative on a human is the worst available failure mode, and 1.28 s is comfortably inside budget for a gated pipeline. Accuracy wins over a sub-second latency edge.

## 4.2 Why Groq is primary for reasoning

227 ms against 83 s cold start for the same model class on the primary provider, which is additionally rate-limited near 40 RPM. The provider abstraction fails over automatically, so neither is load-bearing alone.

## 4.3 Why not CLIP, and what replaced it

The intended joint image/text encoder returns 404 for developer keys. The working equivalent produces 2048-dimensional vectors with image and text confirmed in the *same* space — verified by embedding a photograph and two text strings:

| Comparison                                              | Cosine  |
|---------------------------------------------------------|---------|
| image of a blue truck ↔ “a blue pickup truck on a road” | +0.3367 |
| image of a blue truck ↔ “a bowl of hot noodle soup”     | +0.1234 |

A margin of +0.2133. That shared space is what makes “a white pickup” retrieve frames whose captions never used those words.

# 5 Results

### Scenario suite

**8 / 8** scenarios correct   **1.00** precision   **1.00** recall   **1.00** F<sub>1</sub>

True-negative scenarios (routine delivery, wildlife at the fence, shift change) are weighted equally in this suite. A security system that cries wolf is switched off, and then protects nothing.

Both outputs the brief names by example are reproduced from the running system: “*Blue Ford F-150 spotted at loading dock, 12:00*” and “*Person loitering at main gate, 00:01*”.

## 5.1 Gate efficiency

**38.9%** of frames never reached a model on real footage, **96.7%** on an idle patrol

### Read these together, not separately

Real-footage figures are a floor, not a typical value: every available clip is a CV demo reel authored for continuous motion, which is close to worst-case for a gate. The idle context is a constructed static scene representing patrol idle time and is labelled as such.

**Table 3:** Scenario suite. Three of the eight expect silence and are weighted equally: a system that alerts on the delivery driver, the stray dog and the shift change is switched off within a week, and then protects nothing.

| Scenario                | Expected                                                  | Raised | Suppressed | Result      |
|-------------------------|-----------------------------------------------------------|--------|------------|-------------|
| delivery-daytime        | \{\}\textit{silence}                                      | 0      | 0          | <b>PASS</b> |
| loiter-midnight         | loitering                                                 | 2      | 0          | <b>PASS</b> |
| fence-breach-night      | after-hours-vehicle,<br>perimeter-breach, restricted-zone | 5      | 0          | <b>PASS</b> |
| repeat-visitor          | after-hours-vehicle,<br>baseline-anomaly                  | 4      | 0          | <b>PASS</b> |
| wildlife-false-positive | \{\}\textit{silence}                                      | 0      | 0          | <b>PASS</b> |
| unattended-package      | unattended-object                                         | 1      | 0          | <b>PASS</b> |
| shift-change-busy       | \{\}\textit{silence}                                      | 0      | 0          | <b>PASS</b> |
| tailgating              | after-hours-vehicle, tailgating                           | 4      | 0          | <b>PASS</b> |

**Table 4:** Gate efficiency by operational context. The gate is context-adaptive by design: it spends more where missing something is expensive.

| Context                             | Analysed | Skipped | Skip rate |
|-------------------------------------|----------|---------|-----------|
| restricted zone, 02:00              | 40       | 20      | 33.3%     |
| substation, 02:00                   | 40       | 20      | 33.3%     |
| yard, 14:00                         | 33       | 27      | 45.0%     |
| staff parking, 14:00                | 33       | 27      | 45.0%     |
| parking, quiet clip, 14:00          | 42       | 18      | 30.0%     |
| yard, quiet clip, 11:00             | 32       | 28      | 46.7%     |
| <i>synthetic-idle (constructed)</i> | 2        | 58      | 96.7%     |

### A prediction that did not survive measurement

The design plan asserted “~94% of frames never reach a VLM”. Measured on real footage it is 38.9%. That figure is reported rather than retuned, alongside the idle-patrol number, because either alone would mislead — every available clip is a computer-vision demo reel authored for *continuous* motion, which is close to worst case for a gate.

## 5.2 Retrieval

**Table 5:** Hybrid retrieval over one ingested session. Indicative rather than a benchmark — four queries, one session.

| Query                             | Plan       | Relevant | P@k  | Latency |
|-----------------------------------|------------|----------|------|---------|
| show me all person events         | structured | 10/10    | 1.00 | 2120 ms |
| anything at the substation        | structured | 4/4      | 1.00 | 2021 ms |
| someone in a high-visibility vest | visual     | 9/10     | 0.90 | 2118 ms |
| activity in the restricted core   | structured | 4/4      | 1.00 | 1882 ms |

**0.97** mean precision@k.

## 5.3 Resilience

**Table 6:** Fault injection. Two cases expect *failure*: a cassette miss must raise rather than silently reaching the network, and a camera ray above the horizon must return no position rather than a plausible wrong one.

| Injected fault          | Required behaviour                          | Result      |
|-------------------------|---------------------------------------------|-------------|
| no API key              | falls back to replay                        | <b>PASS</b> |
| cassette miss in replay | raises rather than silently calling out     | <b>PASS</b> |
| corrupt / noise frame   | gate still returns a verdict                | <b>PASS</b> |
| VLM unavailable         | degrades to a low-confidence graph          | <b>PASS</b> |
| telemetry dropout       | gate handles a null telemetry               | <b>PASS</b> |
| ray above horizon       | returns no position rather than a wrong one | <b>PASS</b> |

**6/6** faults degraded gracefully.

Two of those cases expect *failure*. A cassette miss must raise rather than silently reaching the network — otherwise “runs offline” becomes untrue in exactly the situation where it matters. And a camera ray that never meets the ground must return *no* position; a plausible fabricated coordinate would send a responder somewhere real and wrong.

## 6 Beyond the brief

### 6.1 Alerts you can fly to

An alert that says “person at the main gate” is a notification. Every KESTREL alert carries a dispatch position:

```
[CRITICAL] Presence in a restricted area: orange person at Electrical Substation
```



```

raised      02:11:40   confidence 0.69   zone substation
TARGET      18.760018, 73.862886   +/-4.9 m   (geo-projection, conf 0.92)
DISPATCH   230 m from dock, bearing 47 deg, ETA 39 s
FLY AT      18 m,   geofence_ok=True

```

The *source* travels with the coordinates. A metre-accurate projection and a zone-centroid fallback are both useful; confusing them puts a responder at the wrong end of a yard.

## 6.2 Rules written in English, backtested before they fire

Rules are declarative data, not code, so a language model can author them and the engine can replay them over indexed history *before* they are enabled. The operator sees what a proposed rule *would have done* — a rule that would have fired 400 times yesterday is a bad rule, and they find that out here rather than at 3am.

Rules may also carry a `visual_predicate`: their own detector prompt in plain English, so a rule can detect something no fixed class list covers.

## 6.3 A permission boundary that is code, not prompt

Tools that change state or move an aircraft **cannot be executed by the agent**. Proposal and approval are separate code paths; a source-level test asserts the agent’s own loop cannot even pass the approval flag. An agent that can launch an aircraft needs that enforced by the code path — prompt wording would pass a casual review while being one jailbreak from irrelevance.

## 6.4 Findings only a fleet reveals

*“This vehicle has now been seen at three of your sites in five days.”* That is a reconnaissance pattern, and no amount of analysis at one site reveals it, because the evidence is distributed. Since entity embeddings already exist for local re-identification, comparing them across sites costs almost nothing.

## 6.5 Decisions that can be audited

Every consequential event is appended to a hash chain where each entry commits to its predecessor. Altering any historical row invalidates every hash after it. The verification is exposed as a tool and asserted by a test that deliberately tampers with a row — a tamper-evident log nobody verifies is just a log.

# 7 What I would do with more time

1. **A labelled caption set and a proper VLM evaluation.** Caption quality currently rests on a three-signal probe and inspection. With ground truth I would score faithfulness and hallucination rate per model.
2. **Re-identification against a real benchmark.** The matching logic is tested; its accuracy is not measured. General embeddings are the weakest link in the memory layer.
3. **Calibrated projection with a terrain model.** Flat-ground projection is the largest source of positional error, and a raised loading dock breaks it in a way no test here would catch.
4. **Video-native perception.** Everything is per-frame. A video encoder would capture motion semantics — *climbing*, *running*, *loading* — that per-frame captioning approximates poorly.
5. **Active learning from operator feedback.** Feedback currently adjusts a confidence penalty. It should propose concrete rule amendments and backtest them.

6. **Edge deployment.** Tiers 0–1.5 already run on-device; packaging them for a Jetson-class board with store-and-forward for the cloud tiers is the obvious next step.

## 8 On AI tooling

Claude Code wrote most of the code in this repository. The useful part of that statement is not the volume but the division of labour.

**Where it was decisive:** breadth in one pass; tests written alongside the code because they cost minutes rather than hours; and diagram and prose drafting at production quality.

**Where it was confidently wrong:** it invented a model roster from documentation, four fifths of which was unreachable. It wrote a substring match that silently suppressed every alert containing the word “location” (because “cat” is inside it). It wrote a gate heartbeat that reported *zero* efficiency — the exact inverse of the architecture’s central claim. It asserted a 94% metric that measured 38.9%.

### The division that mattered

The model could not decide *what the product is*, or *what is honest*. It would have reported the flattering gate number and shown sixteen sites without labelling fifteen as simulated. Every instance of restraint here was a human decision — and in every case above the model was fluent and confident, with the correction coming from a probe, a test or a measurement rather than from the model reconsidering.

The practical technique that mattered most: **make the model produce evidence before it produces code**. The probe scripts came first and changed the architecture.

Full account, including all six bugs and how each was caught, in docs/04-ai-tooling-log.md.

## 9 Reproducing every claim

```
uv sync --extra dev                # pinned Python 3.12
uv run kestrel doctor              # environment and model reachability
uv run python scripts/fetch_footage.py  # CC BY 4.0 clips
uv run kestrel ingest --clip worker-zone # a real session
uv run pytest -q                   # 90 tests, offline
uv run python scripts/run_evals.py    # every number in this report
uv run python report/generate_tables.py # regenerate these tables
```

The system runs with **no API key**: cassette-replay mode serves recorded model responses from committed fixtures, so the whole product can be evaluated without signing up for anything.